🧱 AWS ECS (Elastic Container Service)

📘 Table of Contents

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1. 🧠 Overview of ECS

Amazon ECS is a fully managed container orchestration service designed for running Docker containers. It eliminates the need to install and manage your own container orchestration software or hardware.

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2. ⚙️ Core ECS Components

Component	Description
Cluster	A logical grouping of ECS services and tasks.
Task Definition	Blueprint that defines how a container runs (image, CPU, memory, ports).
Task	An instantiation of a task definition.
Service	Manages long-running tasks, load balancing, and scaling.
Container Agent	Software that runs on EC2 instances and communicates with ECS.
Launch Types	Determines how ECS tasks are hosted (Fargate or EC2).

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3. 🛠️ ECS Launch Types: EC2 vs Fargate

Feature	EC2 Launch Type	Fargate Launch Type
Infrastructure	User-managed EC2 instances	AWS-managed serverless compute
Networking	Shared ENI by default	Dedicated ENI per task
Scaling	EC2 Auto Scaling	Task-level auto scaling
Cost Model	Pay per instance	Pay per vCPU and memory usage per task
Use Cases	Custom AMIs, privileged containers	Simpler, isolated apps, microservices

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4. 🌐 ECS Networking Modes

- **Bridge**: Default Docker mode (EC2 only).
- **Host**: Shares host's network stack (EC2 only).
- **None**: No networking.
- **awsvpc**: Assigns an Elastic Network Interface (ENI) to the task — **recommended** for production.

📝 *Note*: Only `awsvpc` mode supports **Fargate**.

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5. 🔐 ECS Security Best Practices

- **IAM Roles for Tasks**: Fine-grained permissions per service.
- **IAM Execution Role**: Used by ECS agent to pull images and send logs.
- **Security Groups**: Attach at ENI level in `awsvpc` mode.
- **Private Subnets**: Run services in private subnets with NAT Gateway.
- **Secrets Management**: Store secrets in **AWS Secrets Manager** or **SSM Parameter Store**.
- **Container Isolation**: Each Fargate task gets its own kernel.

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6. 📊 Monitoring and Logging

Logs

- Use **CloudWatch Logs** with `awslogs` log driver.
- Optional: Forward logs to ELK, OpenSearch, or third-party services.

Metrics

- **CloudWatch Metrics**: CPU, memory usage.
- **Container Insights**: CPU, memory, network, storage metrics per container.

Tracing

- **AWS X-Ray**: Instrument services for tracing.
- ECS supports X-Ray daemon as a sidecar container.

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7. 🛠️ CI/CD Pipeline with ECS

Typical CI/CD Flow:

1. Developer pushes code to **GitHub/CodeCommit**.
2. **CodeBuild** builds Docker image.
3. Image pushed to **ECR**.
4. ECS service is updated via **CodePipeline**.

Tools:

- **CodePipeline**: Orchestrate deployments.
- **CodeBuild**: Build Docker containers.
- **CodeDeploy**: Deploy ECS with Blue/Green support.

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8. 📈 Scaling Strategies

ECS Service Auto Scaling

- **Target Tracking**: Scale based on CPU/memory thresholds.
- **Step Scaling**: Scale by fixed amount based on metric breach.

Task Scheduling

- Use scheduled tasks (cron) via **EventBridge**.

Spot Capacity

- Use **Fargate Spot** or **EC2 Spot Instances** for batch jobs and non-critical workloads.

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9. 🚀 Deployment Strategies

Rolling Updates (default)

- ECS replaces tasks in-place.

Blue/Green Deployments

- Use **CodeDeploy** with ECS.
- Shift traffic using two **Target Groups** behind an ALB.
- Supports linear, canary, and all-at-once rollout.

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10. 💰 Cost Optimization

- Use **Fargate Spot** for dev/test workloads.
- Turn off idle ECS tasks or EC2 instances.
- Rightsize CPU/memory in task definitions.

- Leverage **Savings Plans** if using Fargate extensively.
- Use **Compute Optimizer** recommendations.

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11. 🧱 Reference Architecture: Web App on ECS Fargate

Components:

- **Frontend**: React SPA on S3 + CloudFront
- **Backend**: ECS Fargate service with ALB
- **Database**: Amazon RDS
- **Secrets**: AWS Secrets Manager
- **Auth**: Amazon Cognito or JWT
- **Monitoring**: CloudWatch + X-Ray

Security:

- HTTPS via ALB
- Private subnets + NAT Gateway
- WAF for ALB

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12. 🔗 Integration with Other AWS Services

Service	Purpose
ECR	Container image storage
CloudWatch	Logs and metrics
App Mesh	Service mesh for inter-service communication

Cloud Map	DNS-based service discovery
CodeBuild/Deploy	CI/CD automation
Secrets Manager	Secure storage of environment secrets
X-Ray	Distributed tracing
S3	Static content or shared data store
EventBridge	Trigger ECS tasks on schedules or events

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13. ✅ Best Practices Checklist

- Use **Fargate** unless you need custom EC2 configurations.
- Use **awsvpc** networking mode for better isolation.
- Use **IAM Roles for Tasks** with least privilege.
- Store secrets in **Secrets Manager** or **SSM**.
- Configure **logging and metrics** to CloudWatch.
- Enable **auto scaling** with proper thresholds.
- Use **Blue/Green deployments** for safer releases.
- Tag ECS resources for cost tracking.
- Use **Cloud Map** or service mesh for discovery.
- Use **multi-AZ** deployment for high availability.
